

Youtube Comment Spam Detection

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Abstract— YouTube is the most commonly used application these days. There are both good users and bad users who use youtube. Spammers have realized that these types of social media are easier to lure the users into malicious activities. In the current project, spam detection was performed on selected youtube comments and we predict the entered comment as ‘SPAM’ or ‘NOT SPAM’. Google Safe Search and Youtube Bookmaker are two tools that can be used to block harmful links; however, they cannot immediately safeguard real-time users. Four Artificial Intelligence approaches have been employed by victimization to run the survey for the spam comment detection methodology: Call tree, Ada Boost, logistic regression, and Random Forest. We shall achieve a 91.65 percent associate degree of accuracy by using neural networks and also to beat the current action course by around 18%. The difficulty of spam and the efficiency of the most well-known AI approaches (SVMs, Bayesian representation, ANNs, k-NN).

Keywords—spam, youtube, social media, comments, predict

1. INTRODUCTION

In the present era, Youtube has become the biggest source of information. Youtube is an open-source application which can be authorized or can be used by each and everyone. There’s almost nothing that cannot be found on Youtube. Every application has its ‘pros and cons’, likewise one of the main cons of Youtube is ‘SPAMMING’ or ‘SPAMMERS’. Spammers are one among the users itself. The other users of youtube are tricked by these spammers. Spamming can be done in different ways such as it can be abuse or trolling in the comments, or it can be some inappropriate comments and some of the other ways are like advertising or like selling things through the comments. It splits the dataset into various grams and then classifies them into different categories [1].

Spamming is undoubtedly risky because it has the ability to put end users' cyber security at risk. The spammers are well aware of the threats and their consequences. So, they take advantage of them and perform some kind of activities such as hijacking the bank information or the information about the credit card. Some of the spammers even spread the malware through the comments which exploits the vulnerabilities in the machine used by the user; this might affect or annoy the youtube user.

One such malware which was spread through the youtube comments was “WANNA CRY ” ransomware. This incident is still considered as one of the most dangerous cyber-attacks. This incident took place on May 12th 2017, it almost affected around 250000 computer systems and demanded ransom from those in the form of digital money such as cryptocurrency bitcoin from about 25 -28 languages. This malware also had an effect on the NHS (National Health Services) which forced the hospitals under them to postpone some of the major operations. But within 4-5 days people were brought to notice that only the information passed through the calls should be taken into consideration and if they had any doctor appointments and consulting can be done as scheduled.

The spammers wantedly target the comment section by posting completely irrelevant comments, messages, advertisements, ideas, and links that lead to many pop-ups etc. Youtube has around more than 10-11 billion users and around 250 million users (clients) generate around 450-500 million new Youtube content (videos) every day. This content might be anything such as gaming, tech-news, news, live streams, music and what not...! So, this is good in one way, but it also opens the door for the spammers to create enthusiasm or excitement for the users to click the links they share through the comment

section which leads them to various malicious sites which contain malware that may cause harm to their device.

There are many studies going on currently regarding these comment spammers. And some of the temporary ways of identifying the spams are by using the SVM's. The main objective of the project is to create a framework for the detection of the Youtube spams and generate some awareness among the users of youtube, so that they can identify the features of spam and spread of malware can be reduced.

2. REVIEW OF LITERATURE RESEARCH:

Currently spamming has become a very vastly used platform for the spread of malware. There are different kinds of spam, such as social media spamming, web spamming, mail spamming, SMS spamming and others. Our project will be mainly focusing on youtube spamming.

2.1 APPROACH TOWARDS SPAM DETECTION:

Youtube is an enormously used social media platform that's open for all, it doesn't categorize the users. It is open for both content creators and the ones who use it for malicious activities like to spread the malware or the one who does their promotional activities or the one who exposes the drawbacks of the spamming. Many ways can be used for the identification or detection of spam like AI (Artificial intelligence), or cryptography or even machine learning. We have referred to two of the authors; Alberto and Manwar who mentioned Machine learning has the capacity to detect the youtube spam. They used SVM (Support Vector Machine) in the classification phase as a classifier. The classification in SVM has binary-to-class. Typically, 0 and 1 are used to represent the classes. The dataset has been divided into two classes, though. As a result, pre-processing is simple, and we can choose features.

Use an n-gram language model to find common online reviews of a particular product. They concentrated on false reviews that could be copied from other ids. All other identical duplicate evaluations were compared to each individual review. Deep learning framework is a conjoint attention mechanism that focuses more on a type of comment and detects it as spam or ham [12]. To predict word sequences, such as which words would come next, N-gram was used. To put it another way, n-grams automatically break down phrases into a number of smaller parts.

A statistical word sequence that represents the probability dispersal of the subsequent disputes based on the outline is referred to as a language model. Evaluations and websites, particularly blog commentaries, explain the context of information. As a result, language representations are employed to recognize junk in text data using semantic analysis.

References describes the anti-social or social performance of individuals and the characteristics of videos which can be utilized to tell spammers from genuine users on YouTube. To determine which operators and whose videos had responsive connections, they first assembled a random sample of users who had uploaded at the designated time. 720 YouTube users participated in the test collection, which involved manually separating spammers from legal users to establish the categories of spam. Irrespective of the topic of a response film, a user is labelled as a spammer if the video contains pornographic content or an advertisement for a specific product. Users reacted to somewhere between 25,000 and 15,000 videos.

Following the application of the SVM classification algorithm in this particular study, spammers might be identified based on social network, user, and video. F-measure, Spam metrics (FP, TN, TP), and accuracy were used to assess the SVM results. By uninterruptedly learning, updating, and calculating how possibly the input information is to be classed as spam, Support Vector Machine is used to categorize data. They discovered user videos on YouTube along with responses to videotapes from social networks[3].

Spam can occasionally include unwanted content with useless information. They typically appear as distracting visual images of engaging information in the form of pictures, words, or videos. There are several studies on spam in literature, including filtering for internet spam, blog spam, email spam, and SMS spam. Unwanted communications are a type of social spam in social networks. The Spam Detection method developed by Alex Gatchalian et al. will encrypt unnecessary and unneeded blog posts, producing significant tales that are far more useful to ongoing stakeholders.

They advocate for an expansion of their efforts to include adversary awareness, online preparedness to change the prediction of art movement remarks, etc., as well as to extend the definition of spam to include URLs, brief removal of messages, etc.

In their study named "A Novel Approach for Youtube Video Spam Detection Victimization," Simran Kanodia et al. To find the YouTube spam comments, "Andrei Markov call Process" uses the Andrei Markov call method. This model provided an accuracy of almost 78.8% [4].

Abdullah. O. Abdullah et.al. The Support Vector Machine model was used in the study titled "A Comparative Analysis of Common YouTube Comment Spam Filtering Techniques" to test and train the datasets. The project's goal is to specify the prediction technique while succinctly introducing machine learning principles. Machine learning will provide a new opportunity to investigate and improve prediction accuracy because it is significantly more advanced than traditional information analysis methods.

2.2 USAGE OF MACHINE LEARNING TECHNIQUES:

The techniques data mining and machine learning, are generally used by those who use it for web mining or text mining. "The information available in the web page or website that can be beneficial or that can be related to the content by heavily trusting on the obtainable machine learning algorithms and other available techniques," according to Liu, is what web mining is. The extraction and classification of entities by means of data mining and machine learning methods is referred to as content mining. Opinion mining is examining the characteristics of a text passage in an effort to determine the sentiment—whether positive, negative, or neutral—of that piece. The classifier analyses the mood of the user and then classify the comment[2].

By examining the text attributes and emotions of different perspectives, a classifier can be taught to categorize new cases. Like opinion mining, review spam detection belongs to the content mining area but also makes use of features that are somewhat related to the content. Using text mining and tongue processing, features are built to explain the review's text. The text representation has been enhanced and the context is filtered with the help of multi-channel spam filtering [5].

The author of the review, the day and time it was submitted, and in which way it varies from other evaluations for the similar good or service are additional features that can be found in reviews. It is crucial to note that, although while the majority of machine learning algorithms now in use aren't as successful as others at detecting review spam, nonetheless they have been shown to be more accurate or dependable than manual detection.

Before we can address the difficulties in enhancing review spam detection, we must first address knowledge collecting. Ensemble learning can be used to filter the comments which enhances the multi word classifications [6]. This method was used by Sun et al to make a review spam dataset.

2.3 FEATURE ENGINEERING FOR REVIEW SPAM DETECTION:

Creation or exploration of features or structures from data is known as feature engineering. We examine and talk about a few of the traits that are typically used to identify review spam. The most well-known of these features, words contained in the assessment's text, have been used in earlier research, as was briefly mentioned in the introduction. The words bag technique, in which features for apiece review contain either discrete words or small clusters of terms discovered in the review's content, is a popular way to achieve this. Additional features of the reviews, reviewers, and items, for example syntactic and verbal characteristics or characteristics that explain reviewer behaviour, have

been employed by studies less frequently. Features that are maintained utilizing the data from a single review are known as review-centric features. The opposite is true for reviewer-centric structures, which take a comprehensive aspect at all of the reviews submitted by a specific author as well as author-specific data.

It is feasible to use various features from a single category, for instance, subsections outlining and illustrating some reviewer- centric and review-centric aspects.

According to Mukherjee et al., Hammad, Li et al., and other researchers, combining more generic features with a bag-of-words is a additional reliable strategy than utilizing the bag-of-words solely. According to an analysis by Mukherjee et al, consuming the unusual behavioural characteristics of the reviewers was more effective than using the verbal characteristics of the assessments themselves. Using Naïve bayes classifier and logistic regression gives us the flexibility of choosing the spam words [7].

We often turn to a lumen approach in order to maintain this coherence [9]. LMs are unit probabilistic techniques used to uncover linguistic possibilities concealed in texts, for instance the likelihood of words or combinations of word within a dialect. Speech recognition, MT, part-of-speech category, parsing, and data recovery are all successfully carried out using LMs. Lumen disagreement techniques have been demonstrated in earlier research to be extremely cost-effective in jobs like prevent journal spam or police investigation nepotistic linkages. As a result, we tend to analyse various sources of data taken from each website in the assortment using an extension of the basic language modelling approach. Another advanced technique that can be used is "Memory efficient text spam detection using brain-inspired hyper dimensional community [8]. The application of a fancy technique for recovering damaged internet links to the detection of online spam is another contribution of this work. It is required to use a variety of pages that have been manually categorized as spam or not spam in sequence to judge the system. These pages are frequently used to train and evaluate the system [10].

3. MATERIALS AND METHODS:

3.1 DATASET DESCRIPTION:

Our dataset consists of different kinds of comments posted in the youtube comment box by different users. Our dataset is extracted from the GitHub website and it consists of around 1970 different comments. For us now the task is to consider all such comments and identify the comments which are not related to the content related to video or if there is any abusive content present in the comment. We also have the liability to change the data in the dataset.

The knowledge that holds within the files are

INDEPENDENT VARIABLES:

Comment_ID: The user ID of the user who has posted the comment.

Author: Username of the user.

Date: The date of the comment was posted.

Content: What is the actual comment posted?

Class: What is the class of the comment (0 or 1).

3.2 ARCHITECTURAL DIAGRAM:

3.3 PREPROCESSING:

Pre-processing of the communications or messages should be done beforehand preparation begins [13-26]. To commence with, all the characters need to be in the form of lowercase. It is best to think of a term that appears in both capital and lowercase letters as a single word rather than as two distinct ones. Every single message in the dataset needs to be tokenized.

3.4 PROPOSED METHOD:

The SVM can instruct the model to look for a hyper plane to divide the assessments of the dataset. It can also be a linear classifier. The uni-gram approach delivers a stronger result when using SVM because it just requires one word. The written word and bi- gram approaches both employ various word combinations. As a result, once planned during a hyperplane, they perplex the classifier and in turn produce less precise results in the comparison to uni-gram method. The analysis likewise reveals that groupings of a uni-gram with a bi-gram and a trigram give results with lower accuracy. Bhavadhaarini A and Beulla Kutty M suggested this approach.

Adaptive boosting, often known as AdaBoost, is a tool that can be used in conjunction with other algorithms. It is a procedure that chooses the features that need to be enhanced to increase the model's capacity for prediction. Simply omitting features that are not important to the model reduces execution time, which improves model performance.

A. DETECTION OF FRAMEWORK:

Feature selection, classification, data collection, and detection are just a few of the phases that make up the framework detection process consuming machine learning practices.

B. DATA COLLECTION:

Collection of data is the basic step of the Machine learning pipeline. As good as the data is; so, will be the predictions made by the model. Various social media platforms, including Youtube, Instagram, Facebook,

Twitter, and Email, are used to collect data during this phase.

C. FEATURE SELECTION:

For class classification, features are chosen. To identify spam in a user's communication, feature classification analyses the content of the message. The features might be anything, including URLs, hashtags, certain keywords, and negative comments, among others. These kinds of datasets are used to train the model and carry out feature selection. Feature combination is a model that combines the comments and segregates them effectively [11].

D. CLASSIFICATION:

The fundamental purpose of classification is to divide the dataset into various classes according to the most appropriate attributes [27-37]. SVM is one such classification method that can be applied to this assignment. SVM is a very effective and productive classification technology that improves our ability to categorize various difficulties. KNN is another straightforward but useful technique that may be applied to classify algorithms.

E. DETECTION:

At last, the results are obtained. The goal of our study is to evaluate the two methods and determine which one is more accurate in identifying spammy YouTube comments.

4. RESULTS AND SCREENSHOTS:

The findings imply that remarkable number of depictions that have been analysed are present and are being used to filter spam on YouTube. In actuality, a sizable portion of them have the option to attain precision rates of 90% with less or even no blocked ham rates. The prediction model follows the classes present in the dataset. There are two classes in the dataset 0 and 1, here 0 indicates the comment is not spam and 1 indicates the comment is spam. As the model is trained with this dataset, the model identifies the similar comments and identifies them as spam or not spam.

II. SCREENSHOTS:



Fig-1 Entered a comment which is spam

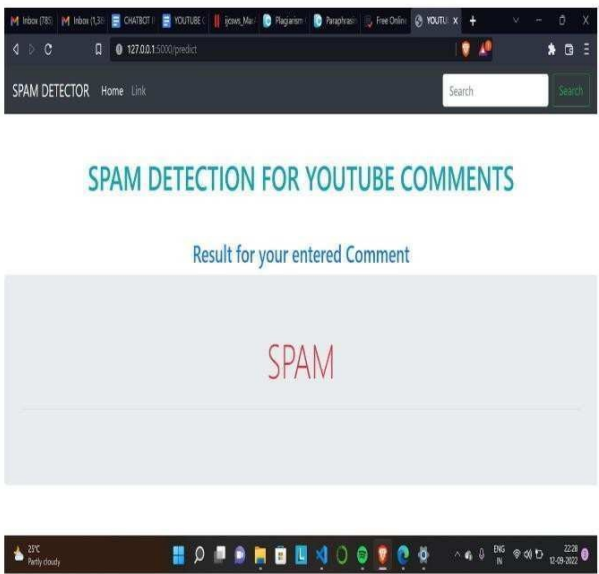


Fig-2 Model detects the entered comment as spam

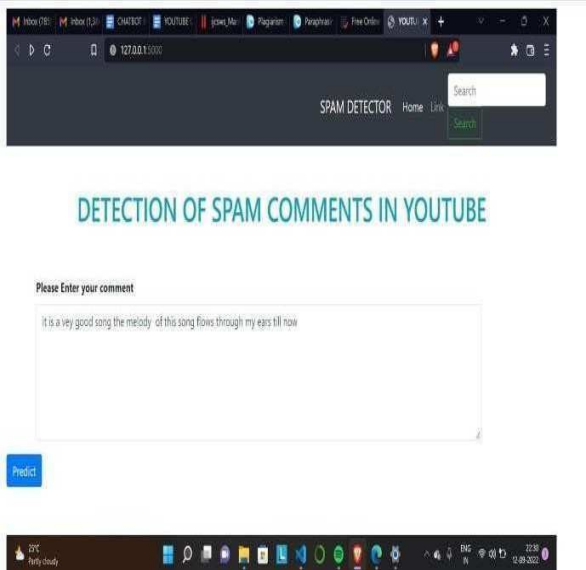


Fig-3 Entered a comment which is not a spam

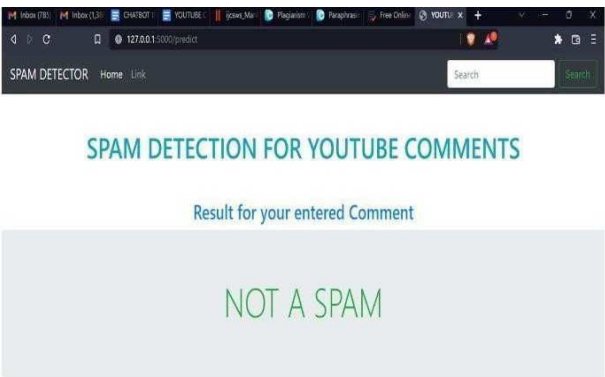


Fig-4 Model identifies the comment as not a spam

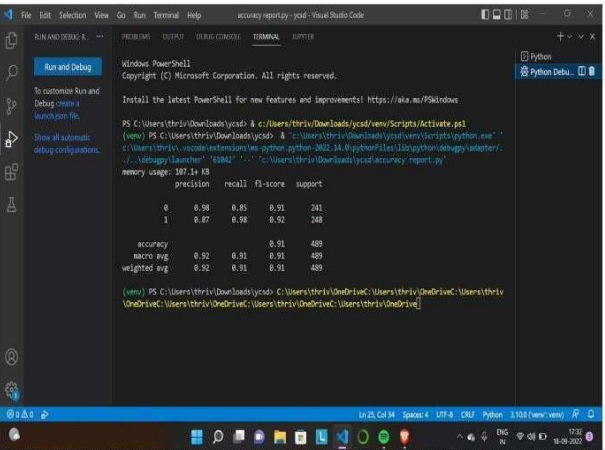


Fig-5 Confusion matrix for our model

5. CONCLUSION:

To categorize the comments as spam or not, we applied various machine learning approaches. We initially trained our model using a dataset, and then we tested it using real-time comments on YouTube. This gave us an accuracy of 93%, which is over 12% higher than the current method. Since YouTube is an open platform for all users, over time the spammers' behaviour may change as a result.

6. REFERENCES:

- [1]. Aiyar, S., & Shetty, N. P. (2018). N-gram assisted youtube spam comment detection. *Procedia computer science*, 132, 174-182.
- [2]. Ezpeleta, E., Iturbe, M., Garitano, I., Mendizabal, I. V. D., & Zurutuza, U. (2018, June). A mood analysis on youtube comments and a method for improved social spam detection. In *International Conference on Hybrid Artificial Intelligence Systems* (pp. 514- 525). Springer, Cham.
- [3]. Ghanem, R., & Erbay, H. (2022). Spam detection on social networks using deep contextualized word representation. *Multimedia Tools and Applications*, 1-16.
- [4]. Yurtseven, İ., Bagriyanik, S., & Ayvaz, S. (2021, September).

- A Review of Spam Detection in social media. In *2021 6th International Conference on Computer Science and Engineering (UBMK)* (pp. 383- 388). IEEE.
- [5]. Novo-Lourés, M., Ruano-Ordás, D., Pavón, R., Laza, R., Gómez-Meire, S., & Méndez, J. R. (2022). Enhancing representation in the context of multiple- channel spam filtering. *Information Processing & Management*, 59(2), 102812.
- [6]. Oh, H. (2022). Corrections to “A YouTube Spam Comments Detection Scheme Using Cascaded Ensemble Machine Learning Model”. *IEEE Access*, 10, 40860-40860.
- [7]. Othman, N. F., & Din, W. I. S. W. (2019). Youtube spam detection framework using naïve bayes and logistic regression. *Indonesian Journal of Electrical Engineering and Computer Science*, 14(3), 1508-1517.
- [8]. Thapa, R., Lamichhane, B., Ma, D., & Jiao, X. (2021, July). Spamhd: Memory- efficient text spam detection using brain-inspired hyperdimensional computing. In *2021 IEEE Computer Society Annual Symposium on VLSI (ISVLSI)* (pp. 84-89). IEEE.
- [9]. Abinaya, R., & Naveen, P. (2020, July). Spam Detection on Social Media Platforms. In *2020 7th International Conference on Smart Structures and Systems (ICSSS)* (pp. 1-3). IEEE.
- [10]. Kanodia, S., Sasheendran, R., & Pathari, V. (2018, September). A novel approach for youtube video spam detection using markov decision process. In *2018 International Conference on Advances in Computing, Communications and Informatics (ICACCI)* (pp. 60-66). IEEE.
- [11]. Li, M., Wu, B., & Wang, Y. (2019, May). Comment spam detection via effective features combination. In *ICC 2019-2019 IEEE International Conference on Communications (ICC)* (pp. 1-6). IEEE.
- [12]. Elakkiya, E., Selvakumar, S., & Leela Velusamy, R. (2021). TextSpamDetector: textual content based deep learning framework for social spam detection using conjoint attention mechanism. *Journal of Ambient Intelligence and Humanized Computing*, 12(10), 9287-9302.
- [13]. Sudar, K. M., Beulah, M., Deepalakshmi, P., Nagaraj, P., and Chinnasamy, P. (2021, January). Detection of Distributed Denial of Service Attacks in SDN using Machine learning techniques. In *2021 International Conference on Computer Communication and Informatics (ICCCI)* (pp. 1-5). IEEE.
- [14]. Sudar, K. M., Deepalakshmi, P., Nagaraj, P., and Muneeswaran, V. (2020, November). Analysis of Cyberattacks and its Detection Mechanisms. In *2020 Fifth International Conference on Research in Computational Intelligence and Communication Networks (ICRCICN)* (pp. 12-16). IEEE.
- [15]. Muneeswaran, V., Nagaraj, P., Godwin, S., Vasundhara, M., and Kalyan, G. (2021, May). Codification of Dental codes for the Cogent Recognition of an Individual. In *2021 5th International Conference on Intelligent Computing and Control Systems (ICICCS)* (pp. 1387-1390). IEEE.
- [16]. Muneeswaran, V., BenSujitha, B., Sujin, B., and Nagaraj, P. (2020). A compendious study on security challenges in big data and approaches of feature selection. *International Journal of Control and Automation*, 13(3), 23-31.
- [17]. Sudar, K. M., Deepalakshmi, P., Ponmozhi, K., and Nagaraj, P. (2019, December). Analysis of Security Threats and Countermeasures for various Biometric Techniques. In *2019 IEEE International Conference on Clean Energy and Energy Efficient Electronics Circuit for Sustainable Development (INCCES)* (pp. 1-6). IEEE.
- [18]. Varma, C. G., Nagaraj, P., Muneeswaran, V., Mokshagni, M., and Jaswanth, M. (2021, May). Astute Segmentation and Classification of leucocytes in blood microscopic smear images using titivated K-means clustering and robust SVM techniques. In *2021 5th International Conference on Intelligent Computing and Control Systems (ICICCS)* (pp. 818-824). IEEE.
- [19]. Anand, S. S., Devaraj, A. F. S., Devi, R. K., Subramanian, C. B., Subramanian, R. R., and Nagaraj, P. (2021). Effective Design and Implementation of B. Tech (CSE) Curriculum with Industry Tie-Ups. *Journal of Engineering Education Transformations*, 34, 191-200.
- [20]. Sudar, K. M., Nagaraj, P., Deepalakshmi, P., and Chinnasamy, P. (2021, January). Analysis of Intruder Detection in Big Data Analytics. In *2021 International Conference on Computer Communication and Informatics (ICCCI)* (pp. 1-5). IEEE.
- [21]. Sudar, K. M., Lokesh, D. L., Chowdary, Y. C., and Chinnasamy, P. (2021, January). Gas Level Detection and Automatic Booking Notification Using IOT. In *2021 International Conference on Computer Communication and Informatics (ICCCI)* (pp. 1-4). IEEE.
- [22]. Nagaraj, P., and Deepalakshmi, P. (2021). Diabetes Prediction Using Enhanced SVM and Deep Neural Network Learning Techniques: An Algorithmic Approach for Early Screening of Diabetes. *International Journal of Healthcare Information Systems and Informatics (IJHISI)*, 16(4), 1-20.
- [23]. Muneeswaran, V., Nagaraj, P., Sai, K. P., Kumar, E. A., and Chanakya, S. R. (2021, August). Enhanced Image Compression Using Fractal and Tree Seed-Bio Inspired Algorithm. In *2021 Second International Conference on Electronics and Sustainable Communication Systems (ICESC)* (pp. 1125-1130). IEEE.
- [24]. Muneeswaran, V., Nagaraj, M. P., Rajasekaran, M. P., Chaithanya, N. S., Babajan, S., and Reddy, S. U. (2021, July). Indigenous Health Tracking Analyzer Using IoT. In *2021 6th International Conference on Communication and Electronics Systems (ICES)* (pp. 530-533). IEEE.
- [25]. Harinath Reddy, C., Koushik Kumar, B. V., Sai Teja Varma, N., Vidya, S., Nagaraj, P., and Muthamil Sudar, K. (2021, May). Risk Prediction of Lung Disease Using Deep Learning Approach. In *International Conference on Image Processing and Capsule Networks* (pp. 462-471). Springer, Cham.
- [26]. Nagaraj, P., Muneeswaran, V., Muthamil Sudar, K., Hammed, S., Lokesh, D. L., and Samara Simha Reddy, V. (2021, May). An Exemplary Template Matching Techniques for Counterfeit Currency Detection. In *International Conference on Image Processing and Capsule Networks* (pp. 370-378). Springer, Cham.
- [27]. Nagaraj, P., Muneeswaran, V., Sudar, K. M., Ali, R. S., Someshwara, A. L., and Kumar, T. S. (2021, May). Internet of Things Based Smart Hospital Saline Monitoring System. In *2021 5th International Conference on Computer, Communication and Signal Processing (ICCCSP)* (pp. 53-58). IEEE.
- [28]. Nagaraj, P., Muneeswaran, V., Pallikonda Rajasekaran, M., Muthamil Sudar, K., and Sumithra, M. (2021). Implementation of Automatic Soil Moisture Dearth Test and Data Exertion Using Internet of Things. In *Emerging Technologies in Data Mining and Information Security* (pp. 511-517). Springer, Singapore.
- [29]. Nagaraj, P., Deepalakshmi, P., and Romany, F. M. (2021). Artificial Flora Algorithm-Based Feature Selection with Gradient Boosted Tree Model for Diabetes Classification. *Diabetes, Metabolic Syndrome and Obesity: Targets and Therapy*, 14, 2789.
- [30]. Muthamil Sudar, K., & Deepalakshmi, P. (2020). A two level security mechanism to detect a DDoS flooding attack in software-defined networks using entropy-based and C4. 5 technique. *Journal of High Speed Networks*, 26(1), 55-76.
- [31]. Muthamil Sudar, K., & Deepalakshmi, P. (2021). An intelligent flow-based and signature-based IDS for SDNs using ensemble feature selection and a multi-layer machine learning-based classifier. *Journal of Intelligent & Fuzzy Systems*, 40(3), 4237-4256.
- [32]. Sudar, K. M., & Deepalakshmi, P. (2020). Comparative study on IDS using machine learning approaches for software defined networks. *International Journal of Intelligent Enterprise*, 7(1-3), 15-27.
- [33]. A, P., & K, V. K. (2022). Convolutional Neural Networks for Medical Image Diagnosis and Prognosis. *International Journal of Computer Communication and Informatics*, 4(2), 54-62.